

The robust identification of exchange from T_2 – T_2 time-domain features



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ABSTRACT

Two-dimensional spin–spin relaxation (T_2 – T_2) techniques have been developed to probe coupling between different environments such as diffusive coupling between small and large pores or chemical exchange with clays. In these studies, Numerical Laplace Inversion (NLI) is used to obtain two-dimensional T_2 – T_2 relaxation distribution spectrum from the T_2 – T_2 signal decays, and the off-diagonal peaks characterize coupling. Often, these coupling peaks are both weak and close to the diagonal and combined with the inherently ill-conditioned nature of the inversion, their presence is difficult to differentiate from inversion related artifacts and blurring. This manuscript presents a time domain based analysis to identify the presence of coupling that avoids the ambiguities present in T_2 – T_2 spectra. The approach utilizes the symmetric nature of the two-dimensional time domain data, where the presence of curvature along one of these symmetries gives an unambiguous indicator of coupling. Measurements on porous glass beads are used to verify the technique.

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1. Introduction

NMR relaxometry yields information about diffusion and molecular dynamics [1–3], and requires minimal field homogeneity unlike conventional NMR spectroscopy and imaging techniques [4]. Relaxometry thus enables a wide range of applications for NMR in small and portable devices. Characterization of NMR T_2 relaxation is a common core component in many applications due to both its natural measurement in any refocusing pulse sequence and broad utility in porous media analysis [5–13]. T_2 measurements are often used to analyze a wide variety of biological [14], geological [15] and synthetic materials [16–18] to estimate quantities such as the pore size distribution and fluid composition (e.g. oil and water content) [3,8,9,19,20]. T_2 relaxation measurements furthermore obtain these quantities in a safe and non-intrusive way [4] both in and outside of the laboratory. For example, a major commercial application of NMR relaxation measurements is to assess the pore size distributions and fluid content of subsurface rock formation in oil well logging [21,22].

Although one-dimensional T_1 and T_2 relaxation distributions can provide the physical properties of porous media, just as in one-dimensional NMR spectroscopy, they provide limited information to resolve multiple components and correlations between

different NMR properties. Multidimensional relaxometry techniques have often proven to be able to address these limitations. Techniques like diffusion– T_2 (D– T_2) [23–25], diffusion–diffusion (D–D) [26,27] and T_1 – T_2 [28,29] correlation techniques have been developed to better analyze pore geometry [27,30,31], identify exchange [1,12,32], and improve fluid identification [20,33,34]. Similar to these multi-dimensional low-field NMR techniques, T_2 can also be correlated to itself. The (T_2 – T_2) relaxation exchange measurement not only includes the information that 1D T_2 measurement offers, but also gives addition information on molecular dynamics, connectivity, as well as T_1 information [21,35]. It has been used to measure the convective flow in partially saturated soil samples [4], violations of the fast diffusion limit [37] and to study diffusional coupling between different pores or in chemical exchange [12,15,35].

For porous media,² the exchange peaks in the T_2 – T_2 spectra can be roughly thought of as the spins that diffuse between different pores. Technically, these peaks correspond to exchange between T_2 relaxation eigenmodes (which are not necessarily exactly the same as pores) and their mutual overlap with the T_1 relaxation eigenmodes [35,36]. This places further requirements on the relaxation properties to observe exchange, such as differing T_1 and T_2 surface relativities and the requirement that one of the pores is not in the fast diffusion limit [35]. Nonetheless, the exchange information T_2 – T_2 contains is not easily available from other multidimensional NMR relaxometry techniques. For instance, while T_1 – T_2 experiments

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² And assuming the absence of coherent motion (e.g. flow, convection, etc.).

contain the same information as T_2 – T_2 experiments [35], exchange information is far more simply shown by and analyzed from cross peaks in the T_2 – T_2 spectra.

T_2 – T_2 measurements of porous media will often detect signals consisting of multiple T_2 decays that correspond to different sample components. Numeric Laplace Inversion (NLI) is necessary to convert this multi-exponential two-dimensional decay data into a T_2 – T_2 spectrum. Then, a sample with a single T_2 relaxation component such as pure water would yield the same exponential decay along each of the two measurement dimensions and would correspond to a single peak along the diagonal of the T_2 – T_2 spectrum. Similarly, a sample with two distinct T_2 relaxation components should result in a T_2 – T_2 spectrum with two separate peaks on the main diagonal line and a pair of off-diagonal peaks arising from exchange between these two components as illustrated in Fig. 1. Saturated porous media typically have even more complex spectra. In natural samples, for instance, soil [21] or rocks, the distributions of pore sizes is rarely mono-disperse and instead form an essentially continuous distribution of pore sizes. Then, the corresponding diagonal T_2 – T_2 peaks will often overlap with each other and form a long continuous peak that reflects the pore size distribution. Furthermore, even in two component systems, the corresponding two diagonal peaks can easily blur together due to inherently limited resolution of NLI. All NLI algorithms must solve an ill-conditioned inversion, which means that a wide range of solutions exist that will fit the data within the noise. While there are a number of regularization techniques [38–41] that can give stable and reasonable spectra, the ill-posed nature of the inversion intrinsically limits the resolution of different T_2 values and large differences in signal strength, and makes the inversion's performance signal to noise ratio (SNR) dependent.

T_2 – T_2 applications in particular frequently push the resolution limits of NLI, where exchange peaks are often weak and adjacent to intense diagonal peaks. Then the optimization of the reconstruction faces the compromise where use of a conservative regularization parameter (allowing for a larger misfit of the spectra to the acquired data) broadens the diagonal-peak to the point of obscuring the presence of cross-peaks, while weakening regularization risks the introduction of artifact peaks that may be mistaken for legitimate cross peaks as the inversion is forced to fit noise in the spectrum [42]. This compromise often makes it difficult to resolve the presence of coupling in T_2 – T_2 correlation maps without unambiguity and worsens with low SNR data [43]. This limitation is intrinsic to T_2 – T_2 correlation maps, limiting the ability of the technique to characterize exchange in cases of strong coupling between distinct T_2 components. In this paper, we introduce a T_2 – T_2 analysis that can unambiguously identify the presence of exchange by directly analyzing the time domain data. In this way, the limitations of NLI that obscure the presence of cross-peaks are avoided to enable the robust identification of exchange.

2. Theory

The Carr–Purcell–Meiboom–Gill (CPMG) is a common pulse sequence [44–46] to measure T_2 relaxation (see Fig. 2a), and forms the building block for the T_2 – T_2 experiment. The CPMG sequence includes a series of refocusing pulse to minimize the additional signal decay due magnetic field inhomogeneity. It starts with a nominal 90° pulse to excite the magnetization into the transverse plane and is followed by a train of 180° pulses to repeatedly re-focus the signal. The spacing time between adjacent 180° pulse is called echo time, t_e , and corresponds to the spacing between the refocused signal. The timing between the initial 90° and 180° pulses is half the echo time and can be improved by shortening the delay to compensate for spin precession during the excitation pulse [47].

The T_2 – T_2 pulse sequence consists of three parts shown in Fig. 2b and essentially consists of two CPMG sequences joined by a mixing period. During the first encoding period τ_1 , any given spin population in a particular chemical environment will have its signal decay as in a 1D CPMG experiment, and its T_2 labels the initial spin population. Typically, no signal is acquired during this period. The second section is the mixing period, and consists of two 90° pulses to form a stimulated echo over the mixing time (τ_m). During the mixing period, the spin populations undergo T_1 relaxation in order to preserve the magnetization and to allow for mixing between different environments (T_2 components). For example, the spins that belong to a spin population A with a relaxation time T_{2a} during τ_1 can diffuse into spin population B where its transverse relaxation time would be T_{2b} . The final spin populations are measured during the third section of the sequence, the detection period (τ_2), as in a standard CPMG acquisition. To form a two-dimensional experiment, for each measurement the mixing time and acquisition times τ_2 are fixed, while only the value of τ_1 is increased by varying the number of echoes during the first period. This time domain data is collected in a two-dimensional matrix and an example of simulated T_2 – T_2 data is shown in Fig. 2d.

Assuming a static sample at equilibrium, the signal with respect to these T_2 periods can be shown to be symmetric. Following, the derivation of the T_2 – T_2 signal in [48] based on the Torrey–Bloch equation, the dynamics of the magnetization during each of the periods of the experiment when undergoing either T_1 and T_2 relaxation can be broken up into the sum of orthogonal relaxation modes $\phi_n(r)$ of magnetization, each with some characteristic decay time T_n .

$$m(r, t) = \sum_{n=0}^{\infty} a_n \phi_n(r) e^{-\frac{t}{T_n}} \quad (1)$$

These are the eigenfunctions and eigenvalues of the Helmholtz equation.

$$\frac{\phi_n(r)}{T_n} + D \nabla^2 \phi_n(r) = 0 \quad (2)$$

with relaxing boundary conditions $Dn \cdot \nabla \phi_n + \rho \phi_n = 0$, where the magnetization loss is proportional to the surface relaxivity ρ .³ In this framework, these relaxation modes are a function of pore geometry and surface relaxivity as these alter the boundary conditions. The relaxation modes for T_1 and T_2 differ when the pores exhibit different ρ where higher relaxivities force faster magnetization loss near the walls and cause greater curvature in the eigenmodes along with shifting the corresponding relaxation values to shorter T_n . Because the modes describing spin relaxation ϕ_n are orthogonal, these transitions can be computed by projecting the different eigenmodes basis onto each other. Thus, the initial state of the system is the projection of the initial magnetization (e.g. uniform after the excitation pulse) projected onto the basis of T_2 relaxation modes: e.g. $a_n = \int \phi_n^*(r) m_i(r, t_i) dv$ where the modes $\phi_n(r)$ have been normalized. Using bra-ket notation to denote these overlap integrals, the magnetization at the start of each section of the T_2 – T_2 experiment is

$$m(r, t) = \sum_{n=0}^{\infty} \phi_n(r) e^{-\frac{t-t_i}{T_n}} \langle \phi_n | m_i \rangle \quad (3)$$

where m_i is the initial magnetization and t_i is the initial time. For the T_2 – T_2 experiment, the initial excitation creates uniform magnetization in the pores ($|i\rangle$) which first undergoes T_2 relaxation ($T_2, |\phi_{2,a}\rangle$) during the preparation period labeled τ_1 . It is then transferred over to longitudinal magnetization during the mixing period where it undergoes T_1 relaxation ($T_1, |\phi_{1,a}\rangle$), labeled τ_m . Finally,

³ The term for bulk relaxation has been omitted, and can be reincorporated by multiplying by the corresponding exponential decay.

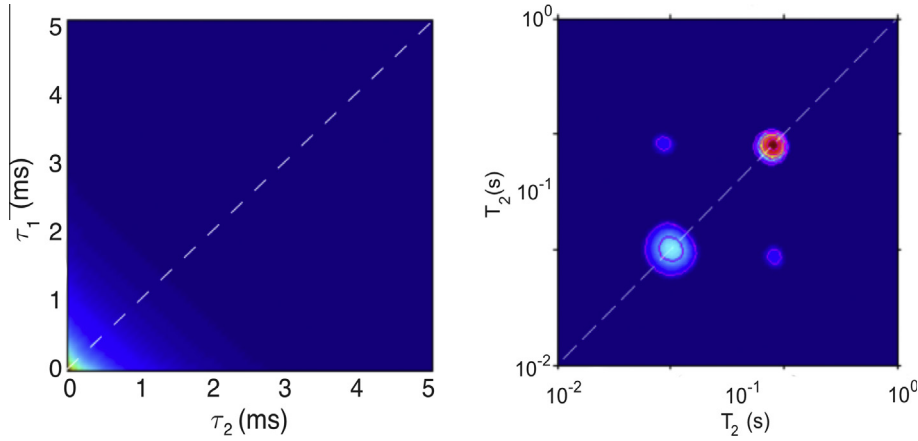


Fig. 1. Simulated T_2 - T_2 time domain data and its relaxation spectrum for two coupled T_2 populations with $T_{2a} = 0.1$ s and $T_{2b} = 1$ s. $t_e = 0.167$ ms.

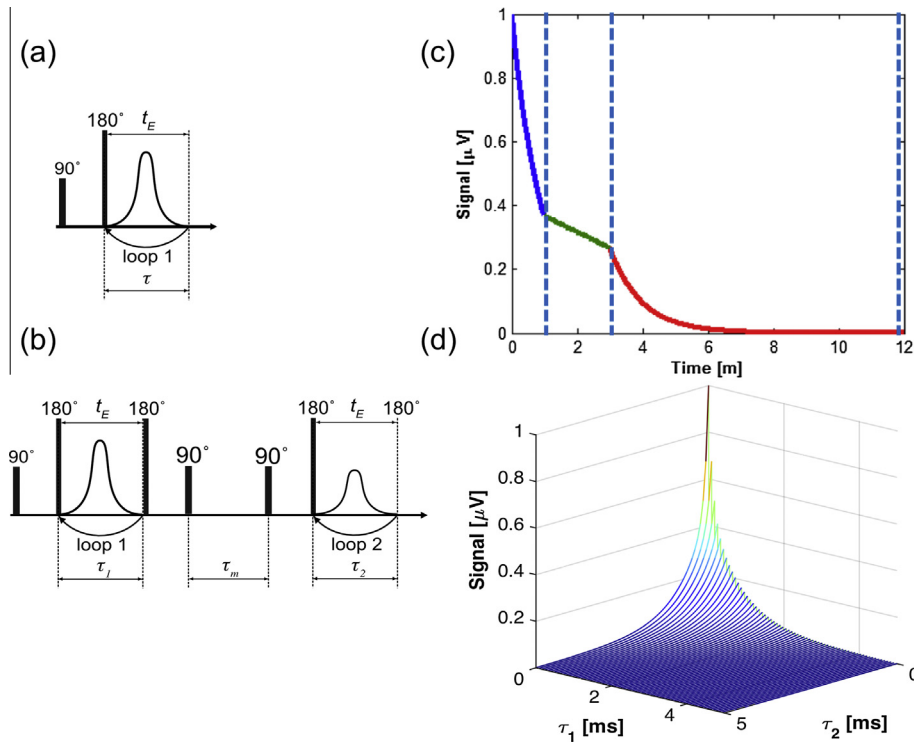


Fig. 2. (a) The CPMG pulse sequence and (b) the T_2 - T_2 pulse sequence. (c) The simulated signal attenuation acquired when collecting the signal for one τ_1 value. The blue line represents the signal undergoes T_2 relaxation in τ_1 period; the green line represents the signal undergoes T_1 relaxation during mixing time period, τ_m ; the red line represents the signal undergoes T_2 relaxation again during τ_2 period where it is detected and aggregated to form a two-dimensional measurement (d) where each line along the τ_2 direction is a single CPMG decay. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

during the detection period it again undergoes T_2 relaxation ($T_2, |\phi_{2,a}\rangle$) where the bulk magnetization ($|f\rangle$) is detected, labeled τ_2 . By repeatedly applying Eq. (3) for each period [43], the detected signal is

$$M(\tau_1^a, \Delta, \tau_2^b) = \sum_{p,q,r} e^{-\frac{\tau_1^b}{T_{2,p}}} e^{-\frac{\tau_1^a}{T_{1,r}}} e^{-\frac{\tau_2^a}{T_{2,p}}} \langle f | \phi_{2,p} \rangle \langle \phi_{2,p} | \phi_{1,r} \rangle \langle \phi_{1,r} | \phi_{2,q} \rangle \langle \phi_{2,q} | i \rangle \quad (4)$$

where the magnetization is the sum over all relaxation modes for each period. As the T_2 modes are identical between the two T_2 periods, $M(\tau_1^b, \Delta, \tau_2^a) = M(\tau_1^a, \Delta, \tau_2^b)$, the signal thus must be symmetric in the time domain data and in the T_2 - T_2 spectra.

The T_2 - T_2 time domain data is simulated/measured in a format of two-dimensional matrix with each dimension corresponding to

the CPMG times in the first and second T_2 encoding periods. This data stored in a two-dimensional matrix $M(\tau_1, \tau_2)$, and is then related to the T_2 - T_2 spectra, according to the exponential kernel applied along the τ_1 and τ_2 directions as shown in Fig. 2d.

$$S_1 = \exp\left(-\frac{\tau_1}{T_{2a}}\right) \quad (5)$$

$$S_2 = \exp\left(-\frac{\tau_2}{T_{2b}}\right) \quad (6)$$

The typical T_2 - T_2 inversion (NLI) of the signal using this kernel does not place the diagonal symmetry constraint on the data, and allows for an asymmetrical T_2 - T_2 spectrum. One way in which this symmetry can be incorporated is by modifying the signal equation to implicitly contain it. The advantage of this approach, over using

a constraint in the inversion, is that this modified signal equation will also reflect these symmetries. The diagonal mirror symmetry of the T_2 - T_2 spectra implies that any peak in spectra (T_{2a}, T_{2b}) must be paired with the same intensity peak mirrored about the diagonal (T_{2b}, T_{2a}) . Thus, the signal equation for the T_2 - T_2 spectrum can be written as the contribution from each pair of cross peaks, instead of as coming from individual values of (T_{2a}, T_{2b}) as shown in Fig. 1. The function for the pair of peaks is:

$$s(\tau_1, \tau_2) = 0.5e^{-\tau_1 R_{2a} - \tau_2 R_{2b}} + 0.5e^{-\tau_1 R_{2b} - \tau_2 R_{2a}} \quad (7)$$

where the relaxation times are represented as rates, $R_{2a} = 1/T_{2a}$, $R_{2b} = 1/T_{2b}$. Note that the on-diagonal peaks are still represented by this model when R_{2a} and R_{2b} are equal. As the principle axes corresponding to the symmetry of the T_2 - T_2 spectrum are along the diagonal $R_s = \frac{1}{2}(R_{2a} + R_{2b})$ and $R_d = \frac{1}{2}(R_{2b} - R_{2a})$, it would also suggest that the symmetry axes of the time domain signal would also correspond to the diagonals $\tau_s = \tau_1 + \tau_2$ and $\tau_d = \tau_2 - \tau_1$, as shown in Fig. 3(a) and (b). Eq. (7) can then be rewritten in these alternative coordinates, for a simplified expression

$$s(\tau_s, \tau_d) = e^{-R_s \tau_s} \times \cosh(\tau_d R_d) \quad (8)$$

This new kernel function is separable into two terms, one is an exponential decay in terms of τ_s and R_s , and the other one is a hyperbolic cosine function in terms of τ_d and R_d . Note that this simple decomposition is in terms of rates and that the corresponding decay time ($\frac{1}{R_{s,d}} \equiv R_{s,d}$) are not the simple sums and differences of the original T_2 times, but of their reciprocals ($\frac{1}{R_{s,d}} = \frac{1}{R_{2b}} \pm \frac{1}{R_{2a}}$).

Because of this separability, the decay along τ_s gives independent information on the mean relaxation rate $R_s = \frac{1}{2}(R_{2a} + R_{2b})$ of the exchange peaks, while variations along τ_d gives information about their off-diagonal separation $R_d = \frac{1}{2}(R_{2b} - R_{2a})$. The func-

tional form of the cosh function along τ_d gives an unambiguous indicator of exchange (cross-peaks) in the time domain data. Independent of R_d and τ_d , the cosh function (and any sum of them given only positive peaks), must increase with positive curvature as τ_d increases. The only exception is in the case that there is no exchange ($R_d = 0$) when the signal contribution is constant with respect to τ_d . Thus, if the acquired signal has curvature along τ_d , there must be legitimate off-diagonal components in the T_2 - T_2 spectrum. To illustrate, simulated data is plotted onto τ_s and τ_d direction (shown in Fig. 3(c) and (d)). In the τ_s slice at $\tau_d = 0$, R_s reflects the total magnetization decay, whereas the curvature along the τ_d cross-section reflects how far the cross peaks are away from each other. In the cross-sections in Fig. 3(c), the deeper curvature of the *cosh* function represents for larger R_d value and flatter curvature corresponding to a small R_d values. If R_d equals to zero, these two cross peaks in spectrum will be on top of each other and consists of a flat line in τ_d cross-sections.

3. Methods

Both simulation data and experimental data are used to verify the signal symmetry and time domain analysis. All the simulation data are calculated using Matlab® with two-site exchange kinetic model [43]. The two site kinetic model, instead of the full model [48], is selected for the simplified calculation of idealized T_2 - T_2 data. Our use of the kinetic model is limited to two sites because its implementation in [43] allows for asymmetry given greater than two sites. This asymmetry is possible in their model for three or more sites because it only assumes detailed mass balance. This looser prior allows for non-equilibria phenomena such as convection and flow that is not typical when analyzing most static porous media samples.

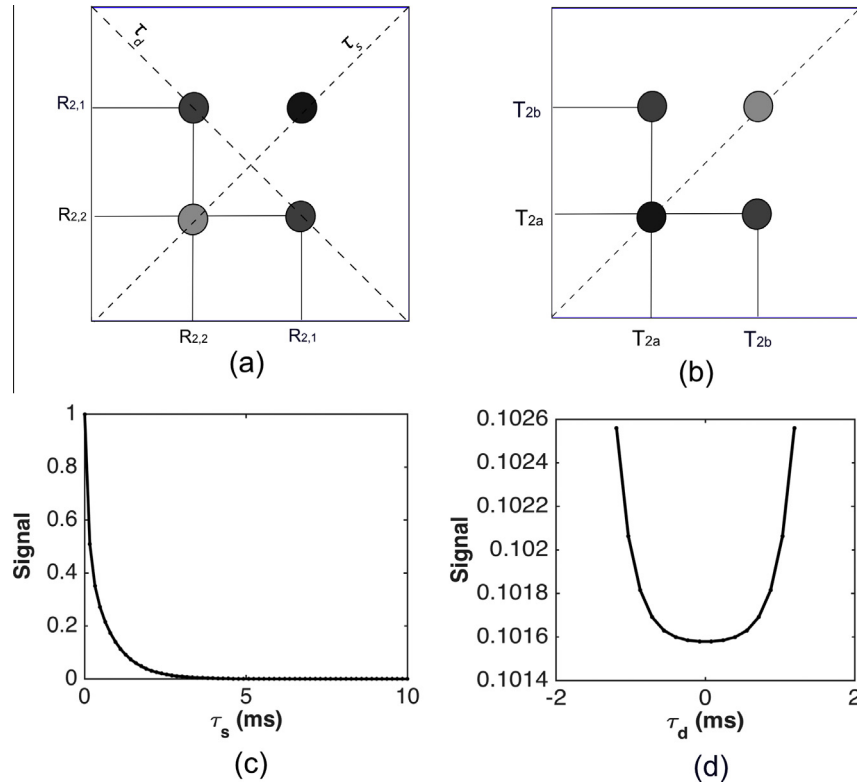


Fig. 3. (a) Schematic T_2 - T_2 spectrum of two coupled spin populations plotted according to the relaxation rates where $R_{2,1} = 1/T_{2a}$ and $R_{2,2} = 1/T_{2b}$. (b) Traditional T_2 - T_2 spectra. (c) Two site exchange simulation data, for the τ_s cross section at $\tau_d = 0$ ms. (d) τ_d cross section at $\tau_s = 1.2698$ ms. Note, the signal axis in (d) has been zoomed in for visualization of the curvature.

For the two site kinetic model, the initial magnetization can be represented as:

$$\mathbf{M} = \begin{bmatrix} M_a \\ M_b \end{bmatrix} \quad (9)$$

where M_a and M_b represent magnetization of the two T_2 populations. The relaxation matrix is:

$$\mathbf{R}_1 = \begin{bmatrix} 1/T_{1a} & 0 \\ 0 & 1/T_{1b} \end{bmatrix} \quad (10)$$

$$\mathbf{R}_2 = \begin{bmatrix} 1/T_{2a} & 0 \\ 0 & 1/T_{2b} \end{bmatrix} \quad (11)$$

And the kinetic matrix represents the exchange rate between spin pool A and spin pool B.

$$\mathbf{K} = \begin{bmatrix} k_a & -k_{ba} \\ -k_{ab} & k_b \end{bmatrix} \quad (12)$$

where k_a is the rate of spins leave and re-enter spin pool T_{2a} and k_{ab} is the exchange rate of spins go from pool A to pool B. The NMR signal is the superposition of all the magnetization components:

$$s(t) = \sum_i [M_i(t) - M_i^{eq}] \quad (13)$$

where M^{eq} is the initial equilibrium magnetization and $M_i(t)$ is the magnetization at time t . The magnetization follows the differential equation:

$$\frac{d}{dt} [\mathbf{M}(t) - \mathbf{M}^{eq}] = -(\mathbf{R} + \mathbf{K})[\mathbf{M}(t) - \mathbf{M}^{eq}] \quad (14)$$

where $\mathbf{R}$ is the relaxation matrix and $\mathbf{K}$ is the kinetic matrix describe the exchange frequencies between different spin pools. The signal adopts T_2 relaxation process, therefore $R_i = 1/T_{2i}$. The simulated time domain data, shown in Figs. 1 and 5, is calculated using Eq. (14) [43].

NMR measurements are performed on a 2 MHz NMR Rock Core Analyzer from Magritek®. For the T_2 - T_2 measurements, τ_2 and τ_1 have the same maximum value, 684.032 ms, with signal being recorded with linear spacing along the τ_2 direction with 4096 echoes and echo time equals to 0.167 ms and signal being recorded with logarithmic spacing along the τ_1 direction for 29 measurements. For each sample, three T_2 - T_2 experiments are done with the mixing times $\tau_m = 0, 5, 50, 100$ ms. For the exchange sample, 150 μm diameter porous glass beads were obtained from Milipore® with 3 μm diameter pores. The pore size and grain size were selected so to produce cross peaks in the T_2 - T_2 map. Two control samples were also used: (1) 150 μm diameter non-porous glass beads and (2) two simultaneously measured glass vials of deionized water with different concentrations of NiCl doping for T_2 relaxation times of 118 ms and 450 ms respectively. All glass beads samples are loose packed, settled by tapping the vial, and vacuum saturated with deionized water.

4. Results and discussion

A set of T_2 - T_2 measurements are performed on porous glass beads, where we expect to observe exchange between the inter- and intra-bead pores. In Fig. 4, the time domain data, T_2 - T_2 spectra and their τ_d slices are compared as the mixing time is changed over 0, 5, 50, and 100 ms. In the T_2 - T_2 spectrum (Fig. 4a), a narrow diagonal line indicates that there is no exchange and is consistent with the fact that a zero mixing time will not have time for exchange to occur. As the mixing time is increased the diagonal peak broadens, this is consistent with increasing exchange but it is also consistent with resolution loss due to the lower SNR at longer mixing times

due to T_1 relaxation and the SNR resolution dependence of NLI. In the two-dimensional time domain data (Fig. 4b), the red contour lines also evolve with mixing time, starting with no variation along τ_d , then increasing in curvature with longer mixing times. The evolution of the curvature in time domain data is clearly observed for the τ_d slice at $\tau_s = 85.04$ ms as shown in Fig. 4c. At zero mixing time, where exchange is impossible, the τ_d line is essentially flat, and at the 100 ms mixing time, the τ_d line curves indicating the presence of exchange.

The advantage of the time domain analysis (Fig. 4b and c) is that the indicator for exchange, curvature, cannot arise from noise. In contrast, the indication of exchange from broadening in the T_2 - T_2 spectra (Fig. 4a) can also occur due to greater regularization in the inversion to compensate for worsening SNR at longer mixing times.

A second feature of the T_2 - T_2 signal decay (Eq. (8)) is that, in the absence of exchange, the decay signal should purely be a function of τ_s , independent of τ_d , and should match the ordinary CPMG decay. A comparison between the first CPMG curve in T_2 - T_2 time domain data (τ_2 curve with $\tau_1 = 0.167$ ms) and the decay along τ_s diagonal ($\tau_d = 0$) are plotted in Fig. 5b. In Fig. 5a, the blue and the red lines, respectively illustrate the locations of these decay curves along τ_2 and τ_s . Along the τ_2 curve $\tau_s \approx \tau_2$ since τ_1 is small (0.167 ms or one echo). As predicted from the model (Eq. (8)), the decay curves show a very good agreement between CPMG decay and τ_2 slice plots (differing by less than the single echo noise amplitude) in the absence of exchange at zero mixing time.

Finally, to confirm that the curvature is indeed due to mixing between T_2 modes and not due to sequence artifacts at non-zero mixing times, the measurements are repeated for two control samples where mixing is impossible. In one control experiment, the sample consists of two separate bottles of NiCl doped water and so exchange between them is impossible. In the second, the sample consists of a non-porous mono-disperse glass-bead pack. Because it has only a single pore size, due to random packing of mono-disperse glass-beads, there are no other T_2 components to exchange with. In Fig. 6, the τ_d cross-sections of the time domain data are plotted and compared to the porous glass beads sample. Consistent with the absence of exchange, there is no significant curvature observed for either sample.

In all of these experiments, the τ_s - τ_d time domain analysis provides a clear, distinct indicator to the presence of exchange peaks in T_2 - T_2 data, and so provides a valuable complement to the analysis of the T_2 - T_2 relaxation spectra. This indicator appears regardless of the number of exchange peaks, but is only sensitive to their presence. By directly analyzing the time domain data, one avoids issues such as peak broadening and artifact peaks from NLI that can easily interfere with the identification of exchange peaks. Thus, the time domain analysis can act as a valuable tool to identify the presence of exchange and guide the subsequent interpretation of the T_2 - T_2 spectra. Application of the time-domain analysis however must also consider that, although the experimental data here agree with the modified kernel function (Eq. (8)), the method relies on the T_2 - T_2 signal having diagonal mirror symmetry and this property presumes a system in equilibrium. Otherwise, signal asymmetry may be possible as shown in recent work that assumes a weaker prior, mass-balance [43]. For characterizing most static samples, equilibrium is nonetheless a fair assumption and can be ensured by appropriate experimental design. The time-domain analysis may only not be potentially applicable in measurements dynamic phenomena such as flow and monitoring chemical reactions, and deviations in T_2 - T_2 signal symmetry can then be used as an indicator for non-equilibrium conditions.

Finally, as presented we only use a simple visual analysis to determine the presence of curvature and identify exchange. Quan-

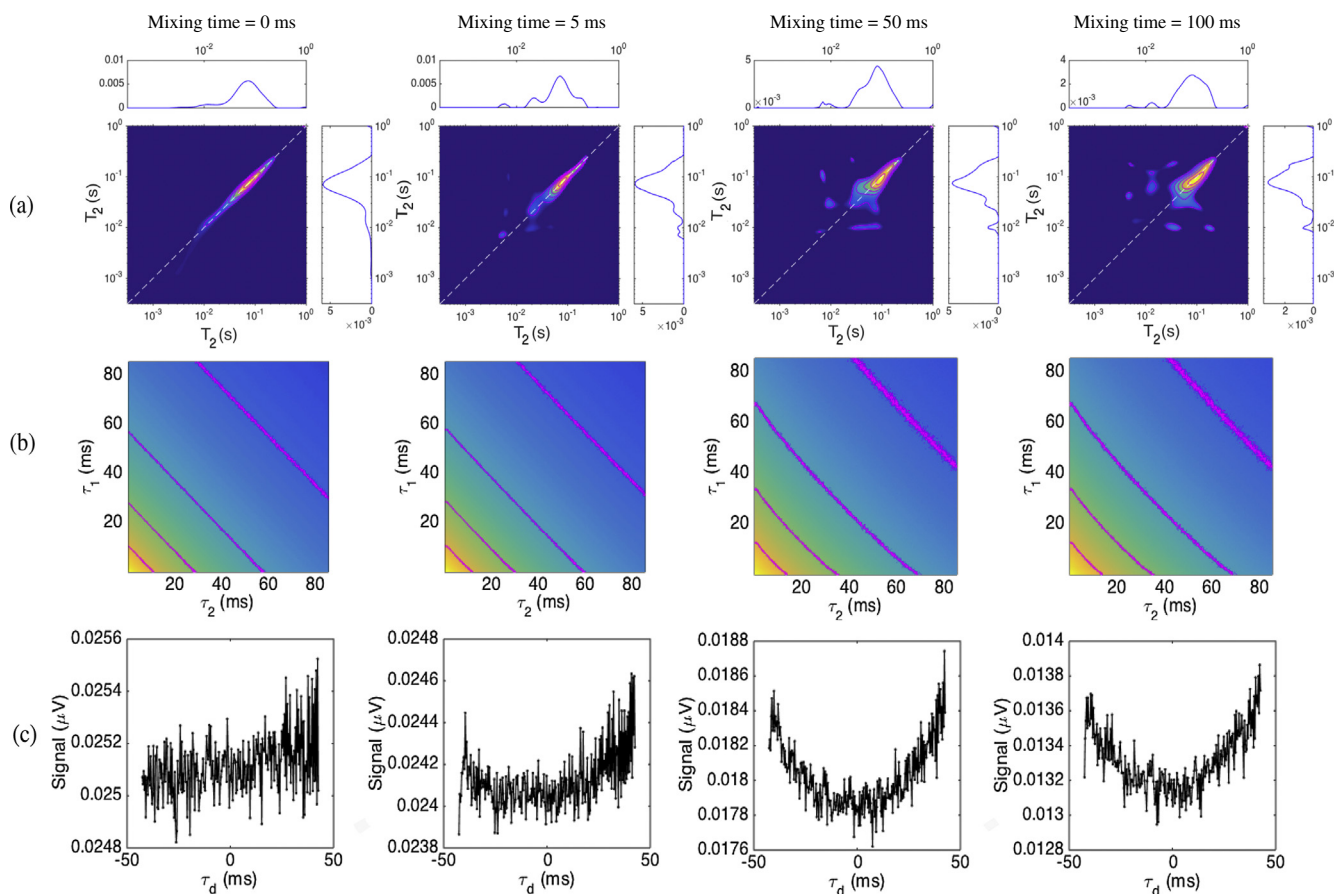


Fig. 4. (a) Numerical Laplace Inversion of T_2 – T_2 time domain data of a porous glass bead pack. All the experimental data presented in the figures above is logarithmic sampled in τ_1 . (b) T_2 – T_2 time domain data of porous glass beads, contour lines are marked red. Note the time domain data is zoomed for early experimental time for better signal intensity. (c) τ_d projection of T_2 – T_2 time domain data with $\tau_s = 85.04$ ms. The signal axes in (c) are zoomed to the signal in order to illustrate the curvature. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

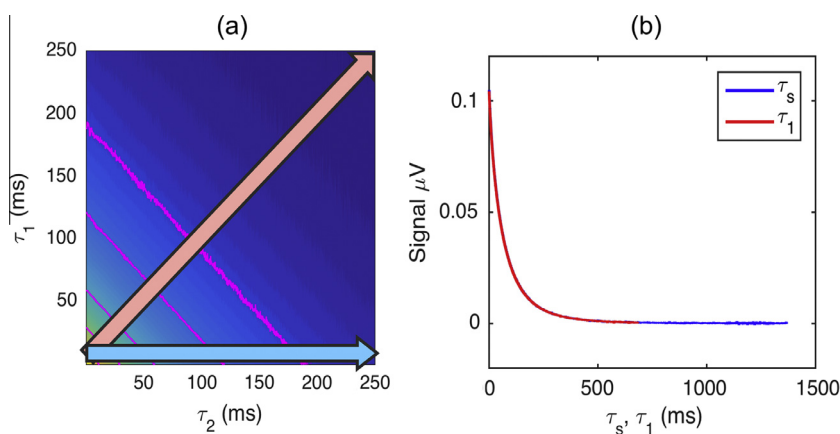


Fig. 5. (a) T_2 – T_2 time domain data of porous glass beads at zero mixing time. The blue arrow indicates the first CPMG curve in the data matrix ($\tau_1 = 0.167$ ms) and red arrow indicates the location of the τ_s signal decay at $\tau_d = 0$. Note: The figure is zoomed in for better signal intensity at early τ_1/τ_2 time. The corresponding signals are plotted in (b). The blue and red curves correspond to the blue and red arrows and the overlap is consistent with the absence of exchange at zero mixing time. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

tification of the curvature should lead to further improvements. Applying a statistical test for curvature should greatly increase the analysis' sensitivity to identifying the presence exchange over a visual analysis, while fitting for the curvature may enable the estimation of cross peak position and area. Significant deviations

from a cosh functional form could also potentially flag the presence of multiple cross-peaks. Furthermore, our symmetric decomposition and alternative description of the T_2 decays as T_s – T_d or R_s – R_d may have additional applications when generating T_2 – T_2 spectra. The spectra from the T_s – T_d or R_s – R_d perspective may have

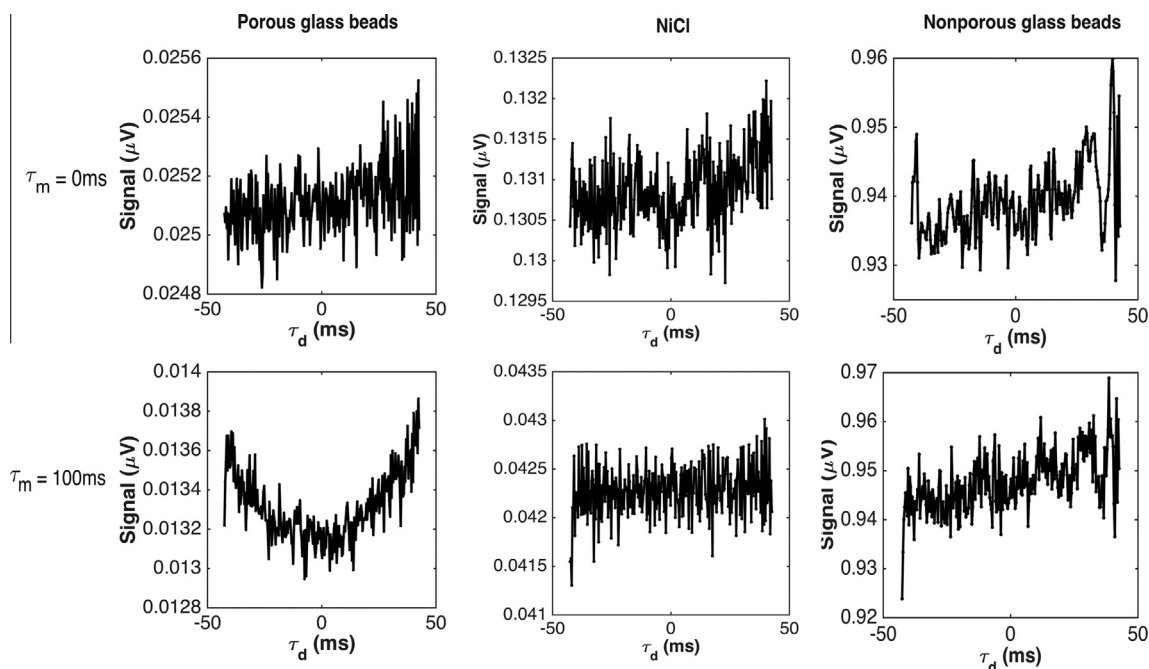


Fig. 6. τ_d projection of time domain data at $\tau_s = 42.75$. Note, all of the signal presented are normalized to the average of first 5 echoes at $\tau_1 = 0$. The signal axes have been zoomed into visualize the curvature.

advantages for interpretation, and its implementation would improve the resolution of the NLI by inherently constraining solutions to incorporate the T_2 – T_2 symmetry.

5. Conclusion

T_2 – T_2 experiments have the valuable capacity to infer exchange, providing information on molecular dynamics, connectivity and diffusional coupling between different pores. However, its traditional analysis requires that the T_2 – T_2 signal decay to first be converted into a two-dimensional T_2 – T_2 relaxation spectrum, an inherently ill-conditioned operation that can frequently hinder the unambiguous identification of exchange peaks. Instead, a direct analysis of the time domain signal can avoid inversion so to directly identify the presence of exchange. Curvature in the signal, when plotted as a function of the difference of the observation and encoding CPMG times ($\tau_d = \tau_2 - \tau_1$), provides an unambiguous indicator of exchange independent of the number of exchanging T_2 populations. This analysis also has the potential to improve quantification of exchange by directly fitting for off-diagonal peak area and position. The method, when applied to porous glass-bead samples, yields a clear indicator of the presence exchange even when the corresponding T_2 – T_2 spectral interpretations are ambiguous and yield only a moderate widening of the on-diagonal non-exchange peak that could also be due variations in the reconstruction parameters. Thus, we expect this time domain analysis to be a valuable tool to inform the T_2 – T_2 spectral analysis, and improve the T_2 – T_2 measurement's sensitivity to exchange.

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